DDR / LPDDR Memory Controller

AXI Interface — Complete Signal Reference

All generations: DDR1 – DDR5 and LPDDR1 – LPDDR5X ● AXI version per generation ● Complete per-channel signal listing with bit widths

Generation	AXI Version	Data Width	Addr Bits	ID Width	Outstanding	Speed
DDR1	AXI3 / AXI4-Lite	32 / 64-bit	32-bit	4-bit	4–8	200–400 MT/s
DDR2	AXI3 / AXI4	32 / 64-bit	32-bit	4-bit	8–16	400–1066 MT/s
DDR3	AXI4	64 / 128-bit	32-bit	4–6-bit	16–32	800–2133 MT/s
DDR4	AXI4	256-bit	36-bit	6–8-bit	32–64	1600–3200 MT/s
DDR5	AXI4 / AXI5	512-bit	40-bit	8-bit	64–256	3200–6400+ MT/s
LPDDR1	AXI3 / proprietary	16 / 32-bit	32-bit	2–4-bit	2–4	200–400 MT/s
LPDDR2	AXI3 / AXI4	32 / 64-bit	32-bit	4-bit	8–16	333–1066 MT/s
LPDDR3	AXI4	64 / 128-bit	32-bit	4–6-bit	16–32	800–2133 MT/s
LPDDR4	AXI4	256-bit	34-bit	6–8-bit	32–64	1600–3200 MT/s
LPDDR4X	AXI4	256-bit	34-bit	8-bit	64	3200–4267 MT/s
LPDDR5	AXI4 / AXI5	512-bit	36-bit	8-bit	128–256	5500–6400 MT/s
LPDDR5X	AXI4 / AXI5	512-bit	38-bit	8-bit	256	7500–8533 MT/s

Legend: Blue rows = DDR family ● Green rows = LPDDR family ● AW = Write Address ● W = Write Data ● B = Write Response ● AR = Read Address ● R = Read Data ● USER signals carry ECC/RAS/traffic-class sideband ● ATOP/STASH = AXI5 optional extensions

PART 1 — DDR SDRAM Generations (DDR1 through DDR5)

DDR1 — DDR SDRAM (JEDEC JESD79) 200–400 MT/s

AXI Version	AXI3 / AXI4-Lite	**Speed**	200–400 MT/s
Data Width	32 / 64-bit	**Addr Bits**	32-bit
ID Width	4-bit	**Outstanding**	4-8

Early embedded MCs used AXI3. Data width matches 32-bit (x4×8 or x8×4) or 64-bit (x8×8) DIMM. Low outstanding depth — row-hit latency ~50 ns was tolerable. AXI3 WID present on W channel.

AW — Write Address Channel

Signal	Bits	Description
AWVALID	1	Master asserts: address is valid
AWREADY	1	MC asserts: ready to accept address
AWID	4	Transaction ID (4 outstanding max)
AWADDR	32	Byte address — 4 GB space
AWLEN	4	Burst length−1 (AXI3: max 16 beats)
AWSIZE	3	Bytes/beat: 010=4B, 011=8B
AWBURST	2	01=INCR (WRAP not commonly used)
AWLOCK	2	AXI3: 2-bit (normal / exclusive / locked)
AWCACHE	4	Memory attributes (cacheable / bufferable)
AWPROT	3	Privilege / security / instruction

W — Write Data Channel

Signal	Bits	Description
WVALID	1	Write data valid
WREADY	1	MC ready to accept data
WID	4	AXI3 write data ID (removed in AXI4)
WDATA	32	Write data (32b for x32 DDR1 module)
WSTRB	4	Byte enables — 1 bit per byte of WDATA
WLAST	1	Last beat of burst

B — Write Response Channel

Signal	Bits	Description
BVALID	1	Response valid
BREADY	1	Master ready
BID	4	Echoes AWID
BRESP	2	00=OKAY 10=SLVERR 11=DECERR

AR — Read Address Channel

Signal	Bits	Description
ARVALID	1	Read address valid
ARREADY	1	MC ready
ARID	4	Read transaction ID
ARADDR	32	Byte address
ARLEN	4	Burst length−1
ARSIZE	3	Bytes per beat
ARBURST	2	Burst type
ARLOCK	2	Lock type (AXI3)
ARCACHE	4	Memory attributes
ARPROT	3	Protection type

R — Read Data Channel

Signal	Bits	Description
RVALID	1	Read data valid
RREADY	1	Master ready to accept
RID	4	Echoes ARID
RDATA	32	Read data
RRESP	2	00=OKAY 10=SLVERR
RLAST	1	Last beat of burst

DDR2 — DDR2 SDRAM (JEDEC JESD79-2) 400–1066 MT/s

AXI Version	AXI3 / AXI4	**Speed**	400–1066 MT/s
Data Width	32 / 64-bit	**Addr Bits**	32-bit
ID Width	4-bit	**Outstanding**	8-16

Higher outstanding depth due to longer latency (OTF latency ∼30–40 cycles). 64-bit data width common for server memory. ODT introduced — MC AXI interface unchanged but DFI gains ODT sideband signals. AXI3 WID still present.

AW — Write Address Channel

Signal	Bits	Description
AWVALID	1	Master asserts: address is valid
AWREADY	1	MC ready
AWID	4	Transaction ID
AWADDR	32	Byte address — 4 GB space
AWLEN	4	Burst length−1 (BL4/BL8 → max 8 beats)
AWSIZE	3	Bytes/beat: 011=8B for x64 bus
AWBURST	2	01=INCR, 10=WRAP (cache use)
AWLOCK	2	AXI3 lock (normal / exclusive / locked)
AWCACHE	4	Cacheable / bufferable attributes
AWPROT	3	Privilege / security

W — Write Data Channel

Signal	Bits	Description
WVALID	1	Write data valid
WREADY	1	MC ready
WID	4	AXI3 write ID
WDATA	64	Write data (64b for x64 module)
WSTRB	8	Byte enables
WLAST	1	Last beat of burst

B — Write Response Channel

Signal	Bits	Description
BVALID	1	Response valid
BREADY	1	Master ready
BID	4	Echoes AWID
BRESP	2	00=OKAY 10=SLVERR 11=DECERR

AR — Read Address Channel

Signal	Bits	Description
ARVALID	1	Read address valid
ARREADY	1	MC ready
ARID	4	Read ID
ARADDR	32	Byte address
ARLEN	4	Burst length−1
ARSIZE	3	Bytes per beat
ARBURST	2	Burst type
ARLOCK	2	Lock type
ARCACHE	4	Memory attributes
ARPROT	3	Protection

R — Read Data Channel

Signal	Bits	Description
RVALID	1	Read data valid
RREADY	1	Master ready
RID	4	Echoes ARID
RDATA	64	Read data
RRESP	2	Response code
RLAST	1	Last beat

DDR3 — DDR3 SDRAM (JEDEC JESD79-3)   800–2133 MT/s

AXI Version	AXI4	**Speed**	800–2133 MT/s
Data Width	64 / 128-bit	**Addr Bits**	32-bit
ID Width	4-6-bit	**Outstanding**	16-32

First widespread AXI4 adoption. BL8 fixed ⇒ AXI data width doubles to 128 b internally to absorb full burst. QoS signals (AWQOS/ARQOS) begin to appear. AXI4 drops WID — write interleaving removed. AWLOCK shrinks to 1-bit.

AW — Write Address Channel

Signal	Bits	Description
AWVALID	1	Master asserts: address is valid
AWREADY	1	MC ready
AWID	6	Transaction ID (up to 64 outstanding)
AWADDR	32	Byte address
AWLEN	8	Burst length−1 (AXI4: up to 256 beats)
AWSIZE	3	Bytes/beat: 100=16B for 128 b bus
AWBURST	2	01=INCR, 10=WRAP
AWLOCK	1	AXI4: 1-bit (normal / exclusive)
AWCACHE	4	Memory attributes
AWPROT	3	Protection type
AWQOS	4	QoS priority 0–15 (first appearance)
AWREGION	4	Logical address region

W — Write Data Channel

Signal	Bits	Description
WVALID	1	Write data valid
WREADY	1	MC ready
WDATA	128	Write data — absorbs full BL8 burst
WSTRB	16	Byte enables (1 per byte of WDATA)
WLAST	1	Last beat (WID removed in AXI4)

B — Write Response Channel

Signal	Bits	Description
BVALID	1	Response valid
BREADY	1	Master ready
BID	6	Echoes AWID
BRESP	2	00=OKAY 10=SLVERR 11=DECERR

AR — Read Address Channel

Signal	Bits	Description
ARVALID	1	Read address valid
ARREADY	1	MC ready
ARID	6	Read ID
ARADDR	32	Byte address
ARLEN	8	Burst length−1
ARSIZE	3	Bytes per beat
ARBURST	2	Burst type
ARLOCK	1	Lock type
ARCACHE	4	Memory attributes
ARPROT	3	Protection
ARQOS	4	QoS priority
ARREGION	4	Address region

R — Read Data Channel

Signal	Bits	Description
RVALID	1	Read data valid
RREADY	1	Master ready
RID	6	Echoes ARID
RDATA	128	Read data
RRESP	2	Response
RLAST	1	Last beat

DDR4 — DDR4 SDRAM (JEDEC JESD79-4) 1600–3200 MT/s

AXI Version	AXI4 with USER signals	**Speed**	1600–3200 MT/s
Data Width	256-bit	**Addr Bits**	36-bit
ID Width	6-8-bit	**Outstanding**	32-64

Data width grows to 256 b matching BL8×64-bit DQ×bank groups. Address expands to 36 b (64 GB). AWUSER/ARUSER appear for traffic class. Multiple AXI ports (2–4) become standard. ECC flags begin appearing in BUSER/RUSER.

AW — Write Address Channel

Signal	Bits	Description
AWVALID	1	Master asserts: address valid
AWREADY	1	MC ready
AWID	8	Transaction ID (256 outstanding)
AWADDR	36	Byte address — 64 GB space
AWLEN	8	Burst length−1
AWSIZE	3	Bytes/beat: 101=32B for 256 b bus
AWBURST	2	01=INCR, 10=WRAP
AWLOCK	1	Exclusive access flag
AWCACHE	4	Memory attributes
AWPROT	3	Protection type
AWQOS	4	QoS: 15=display, 8=CPU, 4=DMA, 0=scrub
AWREGION	4	Logical region
AWUSER	8	Sideband: traffic class, ECC poison tag

W — Write Data Channel

Signal	Bits	Description
WVALID	1	Write data valid
WREADY	1	MC ready
WDATA	256	Write data — full BL8 in one beat
WSTRB	32	Byte enables
WLAST	1	Last beat
WUSER	4	Poison bit, parity hint

B — Write Response Channel

Signal	Bits	Description
BVALID	1	Response valid
BREADY	1	Master ready
BID	8	Echoes AWID
BRESP	2	00=OKAY 10=SLVERR
BUSER	4	ECC writeback status, scrub flag

AR — Read Address Channel

Signal	Bits	Description
`ARVALID`	1	Read address valid
`ARREADY`	1	MC ready
`ARID`	8	Read ID
`ARADDR`	36	Byte address
`ARLEN`	8	Burst length−1
`ARSIZE`	3	Bytes per beat
`ARBURST`	2	Burst type
`ARLOCK`	1	Exclusive flag
`ARCACHE`	4	Memory attributes
`ARPROT`	3	Protection
`ARQOS`	4	QoS priority
`ARREGION`	4	Region
`ARUSER`	8	Traffic class sideband

R — Read Data Channel

Signal	Bits	Description
`RVALID`	1	Read data valid
`RREADY`	1	Master ready
`RID`	8	Echoes ARID
`RDATA`	256	Read data
`RRESP`	2	Response
`RLAST`	1	Last beat
`RUSER`	8	ECC syndrome, corrected-error flag

## DDR5 — DDR5 SDRAM (JEDEC JESD79-5)		3200–6400+ MT/s

AXI Version	AXI4 / AXI5 (emerging)	**Speed**	3200–6400+ MT/s
Data Width	`512-bit`	**Addr Bits**	`40-bit`
ID Width	`8-bit`	**Outstanding**	`64-256`

BL16 + 64 b DQ = 1024 b per burst, delivered as 2×512 b phases. AXI data width = 512 b. Address expands to 40 b (1 TB). Full USER signals used for ECC poison and RAS sideband. Up to 8 AXI ports on modern SoCs. AXI5 AWATOP atomic ops beginning to appear in high-end MC IP.

AW — Write Address Channel

Signal	Bits	Description
`AWVALID`	1	Master asserts: address valid
`AWREADY`	1	MC ready
`AWID`	8	Transaction ID (256 outstanding)
`AWADDR`	40	Byte address — 1 TB space
`AWLEN`	8	Burst length−1 (up to 256 beats)
`AWSIZE`	3	Bytes/beat: `110`=64B for 512 b bus
`AWBURST`	2	`01`=INCR, `10`=WRAP (cache fills)
`AWLOCK`	1	Exclusive access
`AWCACHE`	4	Memory type attributes
`AWPROT`	3	Privilege / security / instruction
`AWQOS`	4	Priority: 15=display, 12=GPU, 8=CPU, 0=ECC
`AWREGION`	4	Address region ID
`AWUSER`	16	ECC poison, RAS tag, traffic class, stash hints
`AWATOP`	6	AXI5: atomic operation type (optional)

W — Write Data Channel

Signal	Bits	Description
`WVALID`	1	Write data valid
`WREADY`	1	MC ready
`WDATA`	512	Write data — one 512 b phase of BL16
`WSTRB`	64	Byte enables (1 bit per byte)
`WLAST`	1	Last beat of burst
`WUSER`	8	ECC poison bit, parity, data check

B — Write Response Channel

Signal	Bits	Description
`BVALID`	1	Response valid
`BREADY`	1	Master ready
`BID`	8	Echoes AWID
`BRESP`	2	`00`=OKAY `10`=SLVERR `11`=DECERR
`BUSER`	4	ECC writeback status, scrub-needed flag

AR — Read Address Channel

Signal	Bits	Description
ARVALID	1	Read address valid
ARREADY	1	MC ready
ARID	8	Read ID
ARADDR	40	Byte address
ARLEN	8	Burst length−1
ARSIZE	3	Bytes per beat
ARBURST	2	Burst type
ARLOCK	1	Exclusive flag
ARCACHE	4	Memory attributes
ARPROT	3	Protection
ARQOS	4	QoS priority
ARREGION	4	Region
ARUSER	16	Traffic class, stash hints, prefetch tags

R — Read Data Channel

Signal	Bits	Description
RVALID	1	Read data valid
RREADY	1	Master ready
RID	8	Echoes ARID
RDATA	512	Read data
RRESP	2	Response code
RLAST	1	Last beat
RUSER	8	ECC syndrome, CE/UE flag, scrub result

PART 2 — LPDDR Generations (LPDDR1 through LPDDR5X)

LPDDR1 — LPDDR (JEDEC JESD209)　200–400 MT/s

AXI Version	AXI3 / proprietary	**Speed**	200–400 MT/s
Data Width	16 / 32-bit	**Addr Bits**	32-bit
ID Width	2-4-bit	**Outstanding**	2-4

Mobile-first MC: extremely low outstanding depth. Power management (CKE gating) exposed via APB/AHB sideband rather than AXI USER signals. 16 b DQ width common for phone SoCs. AXI3 dominant era with WID on W channel.

AW — Write Address Channel

Signal	Bits	Description
AWVALID	1	Address valid
AWREADY	1	MC ready
AWID	2	ID (very small — 4 outstanding max)
AWADDR	32	Byte address
AWLEN	4	Burst length−1 (BL2/BL4/BL8)
AWSIZE	3	Bytes/beat: 001=2B (16 b bus)
AWBURST	2	INCR / WRAP
AWLOCK	2	AXI3 lock
AWCACHE	4	Cache attributes
AWPROT	3	Protection

W — Write Data Channel

Signal	Bits	Description
WVALID	1	Data valid
WREADY	1	MC ready
WID	2	AXI3 write ID
WDATA	16	Write data (16 b for x16 LPDDR)
WSTRB	2	Byte enables
WLAST	1	Last beat

B — Write Response Channel

Signal	Bits	Description
BVALID	1	Response valid
BREADY	1	Master ready
BID	2	Echoes AWID
BRESP	2	Response code

AR — Read Address Channel

Signal	Bits	Description
ARVALID	1	Address valid
ARREADY	1	MC ready
ARID	2	Read ID
ARADDR	32	Address
ARLEN	4	Burst length−1
ARSIZE	3	Bytes/beat
ARBURST	2	Burst type
ARLOCK	2	Lock
ARCACHE	4	Attributes
ARPROT	3	Protection

R — Read Data Channel

Signal	Bits	Description
RVALID	1	Data valid
RREADY	1	Master ready
RID	2	Echoes ARID
RDATA	16	Read data
RRESP	2	Response
RLAST	1	Last beat

LPDDR2 — LPDDR2 (JEDEC JESD209-2) 333–1066 MT/s

AXI Version	AXI3 / AXI4	**Speed**	333–1066 MT/s
Data Width	32 / 64-bit	**Addr Bits**	32-bit
ID Width	4-bit	**Outstanding**	8-16

Introduced CA bus on DRAM side (replaces RAS/CAS/WE) but AXI interface unchanged. 32 b data width for single-channel ×16×2. Power-Down and Self-Refresh now DFI low-power signals. WID still present in AXI3 mode.

AW — Write Address Channel

Signal	Bits	Description
AWVALID	1	Address valid
AWREADY	1	MC ready
AWID	4	Transaction ID
AWADDR	32	Byte address
AWLEN	4	Burst length−1 (BL4/BL8/BL16)
AWSIZE	3	Bytes/beat: 010=4B or 011=8B
AWBURST	2	INCR / WRAP
AWLOCK	2	AXI3 lock type
AWCACHE	4	Memory type
AWPROT	3	Protection

W — Write Data Channel

Signal	Bits	Description
WVALID	1	Data valid
WREADY	1	MC ready
WID	4	AXI3 write ID
WDATA	32	Write data
WSTRB	4	Byte enables
WLAST	1	Last beat

B — Write Response Channel

Signal	Bits	Description
BVALID	1	Response valid
BREADY	1	Master ready
BID	4	Echoes AWID
BRESP	2	Response code

AR — Read Address Channel

Signal	Bits	Description
ARVALID	1	Address valid
ARREADY	1	MC ready
ARID	4	Read ID
ARADDR	32	Address
ARLEN	4	Burst length−1
ARSIZE	3	Bytes/beat
ARBURST	2	Burst type
ARLOCK	2	Lock
ARCACHE	4	Attributes
ARPROT	3	Protection

R — Read Data Channel

Signal	Bits	Description
RVALID	1	Data valid
RREADY	1	Master ready
RID	4	Echoes ARID
RDATA	32	Read data
RRESP	2	Response
RLAST	1	Last beat

LPDDR3 — LPDDR3 (JEDEC JESD209-3) 800–2133 MT/s

AXI Version	AXI4	**Speed**	800–2133 MT/s
Data Width	64 / 128-bit	**Addr Bits**	32-bit
ID Width	4-6-bit	**Outstanding**	16-32

Transition to AXI4; WID removed. BL8 fixed — 64 b data width typical. QoS signals added. PoP (Package-on-Package) placement means MC and PHY may be on different dies. Power management via DFI LP interface, not AXI.

AW — Write Address Channel

Signal	Bits	Description
AWVALID	1	Address valid
AWREADY	1	MC ready
AWID	6	Transaction ID
AWADDR	32	Byte address
AWLEN	8	Burst length−1 (AXI4)
AWSIZE	3	Bytes/beat: 011=8B for 64 b
AWBURST	2	INCR / WRAP
AWLOCK	1	AXI4: 1-bit exclusive
AWCACHE	4	Memory attributes
AWPROT	3	Protection
AWQOS	4	QoS priority
AWREGION	4	Region

W — Write Data Channel

Signal	Bits	Description
WVALID	1	Data valid
WREADY	1	MC ready
WDATA	128	Write data (128 b for BL8×64)
WSTRB	16	Byte enables
WLAST	1	Last beat

B — Write Response Channel

Signal	Bits	Description
BVALID	1	Response valid
BREADY	1	Master ready
BID	6	Echoes AWID
BRESP	2	Response code

AR — Read Address Channel

Signal	Bits	Description
ARVALID	1	Address valid
ARREADY	1	MC ready
ARID	6	Read ID
ARADDR	32	Address
ARLEN	8	Burst length−1
ARSIZE	3	Bytes/beat
ARBURST	2	Burst type
ARLOCK	1	Exclusive
ARCACHE	4	Attributes
ARPROT	3	Protection
ARQOS	4	QoS
ARREGION	4	Region

R — Read Data Channel

Signal	Bits	Description
RVALID	1	Data valid
RREADY	1	Master ready
RID	6	Echoes ARID
RDATA	128	Read data
RRESP	2	Response
RLAST	1	Last beat

LPDDR4 — LPDDR4 (JEDEC JESD209-4) 1600–3200 MT/s

AXI Version	AXI4	**Speed**	1600–3200 MT/s
Data Width	256-bit	**Addr Bits**	34-bit
ID Width	6-8-bit	**Outstanding**	32-64

Dual-channel architecture (2×16 b = 32 b per channel, BL16 = 256 b burst). AWUSER carries channel routing hint. Address expands to 34 b (16 GB). 4 AXI ports common on flagship SoCs.

AW — Write Address Channel

Signal	Bits	Description
AWVALID	1	Address valid
AWREADY	1	MC ready
AWID	8	Transaction ID
AWADDR	34	Byte address — 16 GB space
AWLEN	8	Burst length−1
AWSIZE	3	Bytes/beat: 101=32B for 256 b
AWBURST	2	INCR / WRAP
AWLOCK	1	Exclusive access
AWCACHE	4	Memory attributes
AWPROT	3	Protection
AWQOS	4	QoS: 15=display, 8=CPU, 4=ISP
AWREGION	4	Region
AWUSER	8	Channel hint, traffic class, ECC

W — Write Data Channel

Signal	Bits	Description
WVALID	1	Data valid
WREADY	1	MC ready
WDATA	256	Write data — BL16×32 b channel
WSTRB	32	Byte enables
WLAST	1	Last beat
WUSER	4	ECC poison, data check

B — Write Response Channel

Signal	Bits	Description
BVALID	1	Response valid
BREADY	1	Master ready
BID	8	Echoes AWID
BRESP	2	Response code
BUSER	4	ECC / error status

AR — Read Address Channel

Signal	Bits	Description
ARVALID	1	Address valid
ARREADY	1	MC ready
ARID	8	Read ID
ARADDR	34	Address
ARLEN	8	Burst length−1
ARSIZE	3	Bytes/beat
ARBURST	2	Burst type
ARLOCK	1	Exclusive
ARCACHE	4	Attributes
ARPROT	3	Protection
ARQOS	4	QoS
ARREGION	4	Region
ARUSER	8	Channel routing, traffic class

R — Read Data Channel

Signal	Bits	Description
RVALID	1	Data valid
RREADY	1	Master ready
RID	8	Echoes ARID
RDATA	256	Read data
RRESP	2	Response
RLAST	1	Last beat
RUSER	8	ECC syndrome, error flag

LPDDR4X — LPDDR4X (JEDEC JESD209-4X) 3200–4267 MT/s

AXI Version	AXI4	**Speed**	3200–4267 MT/s
Data Width	256-bit	**Addr Bits**	34-bit
ID Width	8-bit	**Outstanding**	64

AXI interface identical to LPDDR4. Primary change is electrical: lower I/O voltage (0.6 V vs 1.1 V), better power efficiency. PHY-side DFI signals reflect lower swing; AXI layer is unchanged. Most Cortex-A55/A78 cluster MCs use this pairing.

AW — Write Address Channel

Signal	Bits	Description
AWVALID	1	Address valid
AWREADY	1	MC ready
AWID	8	Transaction ID
AWADDR	34	Byte address — 16 GB
AWLEN	8	Burst length−1
AWSIZE	3	Bytes/beat: 101=32B
AWBURST	2	INCR / WRAP
AWLOCK	1	Exclusive
AWCACHE	4	Memory attributes
AWPROT	3	Protection
AWQOS	4	QoS priority
AWREGION	4	Region
AWUSER	8	Traffic class, ECC, channel hint

W — Write Data Channel

Signal	Bits	Description
WVALID	1	Data valid
WREADY	1	MC ready
WDATA	256	Write data (same as LPDDR4)
WSTRB	32	Byte enables
WLAST	1	Last beat
WUSER	4	Poison / parity

B — Write Response Channel

Signal	Bits	Description
BVALID	1	Response valid
BREADY	1	Master ready
BID	8	Echoes AWID
BRESP	2	Response code
BUSER	4	ECC status

AR — Read Address Channel

Signal	Bits	Description
ARVALID	1	Address valid
ARREADY	1	MC ready
ARID	8	Read ID
ARADDR	34	Address
ARLEN	8	Burst length−1
ARSIZE	3	Bytes/beat
ARBURST	2	Burst type
ARLOCK	1	Exclusive
ARCACHE	4	Attributes
ARPROT	3	Protection
ARQOS	4	QoS
ARREGION	4	Region
ARUSER	8	Traffic class, channel hint

R — Read Data Channel

Signal	Bits	Description
RVALID	1	Data valid
RREADY	1	Master ready
RID	8	Echoes ARID
RDATA	256	Read data
RRESP	2	Response
RLAST	1	Last beat
RUSER	8	ECC syndrome / error

LPDDR5 — LPDDR5 (JEDEC JESD209-5) 5500–6400 MT/s

AXI Version	AXI4 / AXI5 (emerging)	**Speed**	5500–6400 MT/s
Data Width	512-bit	**Addr Bits**	36-bit
ID Width	8-bit	**Outstanding**	128-256

BL32 with 32 b per channel × 2 channels = 2048 b per burst, delivered in 512 b phases. WCK (Write Clock) at 2× rate is PHY-internal. Address grows to 36 b (64 GB). AWUSER carries sub-channel routing bit (SC0/SC1) and PPR (Post Package Repair) hints.

AW — Write Address Channel

Signal	Bits	Description
AWVALID	1	Address valid
AWREADY	1	MC ready
AWID	8	Transaction ID
AWADDR	36	Byte address — 64 GB
AWLEN	8	Burst length−1
AWSIZE	3	Bytes/beat: 110=64B for 512 b
AWBURST	2	INCR / WRAP
AWLOCK	1	Exclusive access
AWCACHE	4	Memory attributes
AWPROT	3	Protection
AWQOS	4	QoS: 15=display, 12=GPU, 8=CPU
AWREGION	4	Region
AWUSER	16	Sub-channel sel, ECC poison, RAS, PPR hint
AWATOP	6	AXI5 atomic op (optional)

W — Write Data Channel

Signal	Bits	Description
WVALID	1	Data valid
WREADY	1	MC ready
WDATA	512	Write data — 512 b phase
WSTRB	64	Byte enables
WLAST	1	Last beat
WUSER	8	ECC poison, parity, data check

B — Write Response Channel

Signal	Bits	Description
BVALID	1	Response valid
BREADY	1	Master ready
BID	8	Echoes AWID
BRESP	2	00=OKAY 10=SLVERR
BUSER	8	ECC status, PPR result, sub-channel info

AR — Read Address Channel

Signal	Bits	Description
ARVALID	1	Address valid
ARREADY	1	MC ready
ARID	8	Read ID
ARADDR	36	Address
ARLEN	8	Burst length−1
ARSIZE	3	Bytes/beat
ARBURST	2	Burst type
ARLOCK	1	Exclusive
ARCACHE	4	Attributes
ARPROT	3	Protection
ARQOS	4	QoS
ARREGION	4	Region
ARUSER	16	Sub-channel, prefetch tags, traffic class

R — Read Data Channel

Signal	Bits	Description
RVALID	1	Data valid
RREADY	1	Master ready
RID	8	Echoes ARID
RDATA	512	Read data
RRESP	2	Response
RLAST	1	Last beat
RUSER	8	ECC CE/UE flag, scrub result, sub-ch

LPDDR5X — LPDDR5X (JEDEC JESD209-5X)   7500–8533 MT/s

AXI Version	AXI4 / AXI5	**Speed**	7500–8533 MT/s
Data Width	512-bit	**Addr Bits**	38-bit
ID Width	8-bit	**Outstanding**	256

Address grows to 38 b (256 GB, anticipating 16+ GB DRAM stacks). AXI5 stash hints (AWSTASHNID, AWSTASHLPID) appear in high-end AI/ML SoCs to pre-position cache lines near the requesting CPU cluster. Otherwise electrically identical to LPDDR5 from the AXI perspective.

AW — Write Address Channel

Signal	Bits	Description
AWVALID	1	Address valid
AWREADY	1	MC ready
AWID	8	Transaction ID
AWADDR	38	Byte address — 256 GB
AWLEN	8	Burst length−1
AWSIZE	3	Bytes/beat: 110=64B
AWBURST	2	INCR / WRAP
AWLOCK	1	Exclusive
AWCACHE	4	Memory attributes
AWPROT	3	Protection
AWQOS	4	QoS priority
AWREGION	4	Region
AWUSER	16	Sub-channel, RAS, ECC, AI/ML traffic tag
AWATOP	6	AXI5 atomic ops
AWSTASHNID	11	AXI5 stash: target node ID (optional)
AWSTASHLPID	5	AXI5 stash: logical processor ID (optional)

W — Write Data Channel

Signal	Bits	Description
WVALID	1	Data valid
WREADY	1	MC ready
WDATA	512	Write data
WSTRB	64	Byte enables
WLAST	1	Last beat
WUSER	8	ECC poison, parity

B — Write Response Channel

Signal	Bits	Description
BVALID	1	Response valid
BREADY	1	Master ready
BID	8	Echoes AWID
BRESP	2	Response
BUSER	8	ECC result, sub-channel, PPR status

AR — Read Address Channel

Signal	Bits	Description
ARVALID	1	Address valid
ARREADY	1	MC ready
ARID	8	Read ID
ARADDR	38	Address
ARLEN	8	Burst length−1
ARSIZE	3	Bytes/beat
ARBURST	2	Burst type
ARLOCK	1	Exclusive
ARCACHE	4	Attributes
ARPROT	3	Protection
ARQOS	4	QoS
ARREGION	4	Region
ARUSER	16	Sub-ch, prefetch, stash hints

R — Read Data Channel

Signal	Bits	Description
RVALID	1	Data valid
RREADY	1	Master ready
RID	8	Echoes ARID
RDATA	512	Read data
RRESP	2	Response
RLAST	1	Last beat
RUSER	8	ECC CE/UE, scrub, sub-ch info